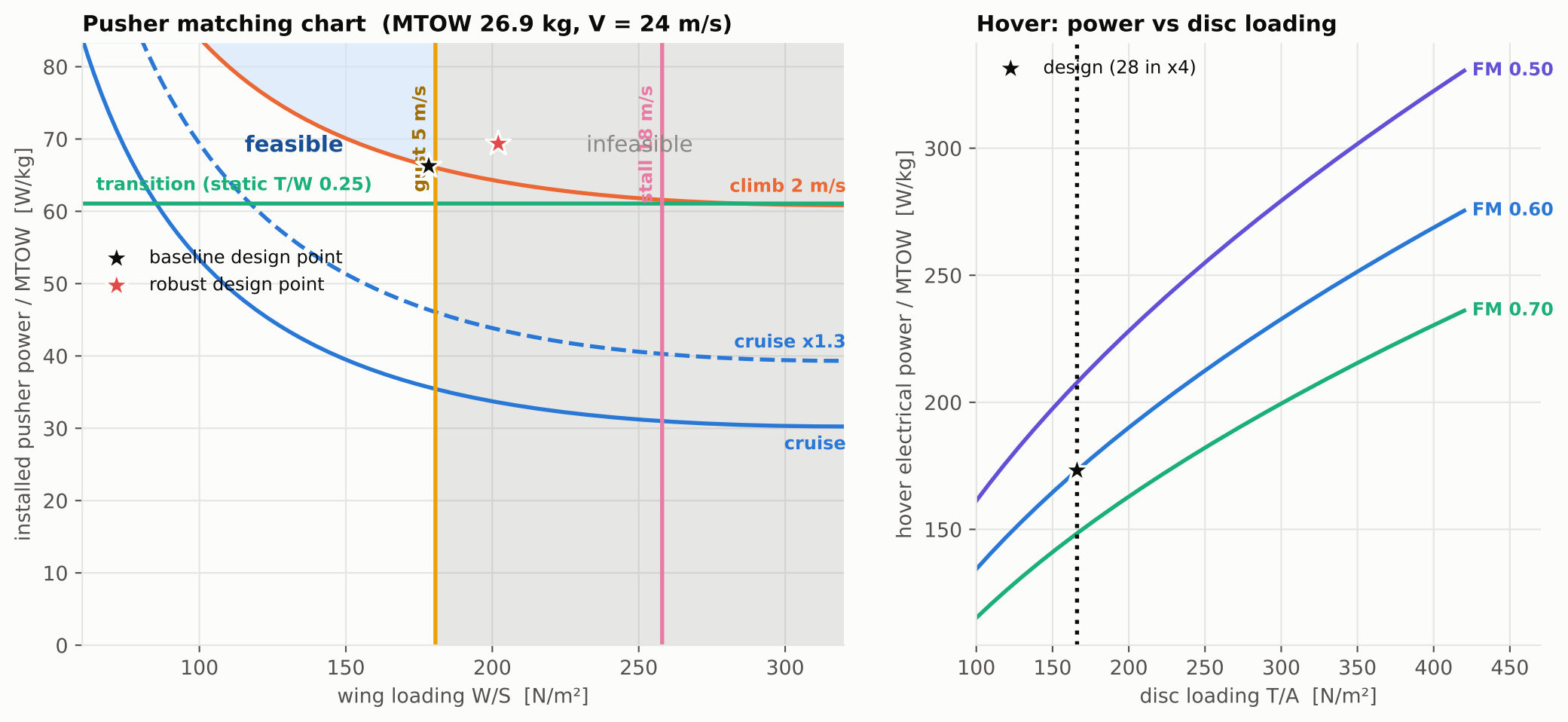
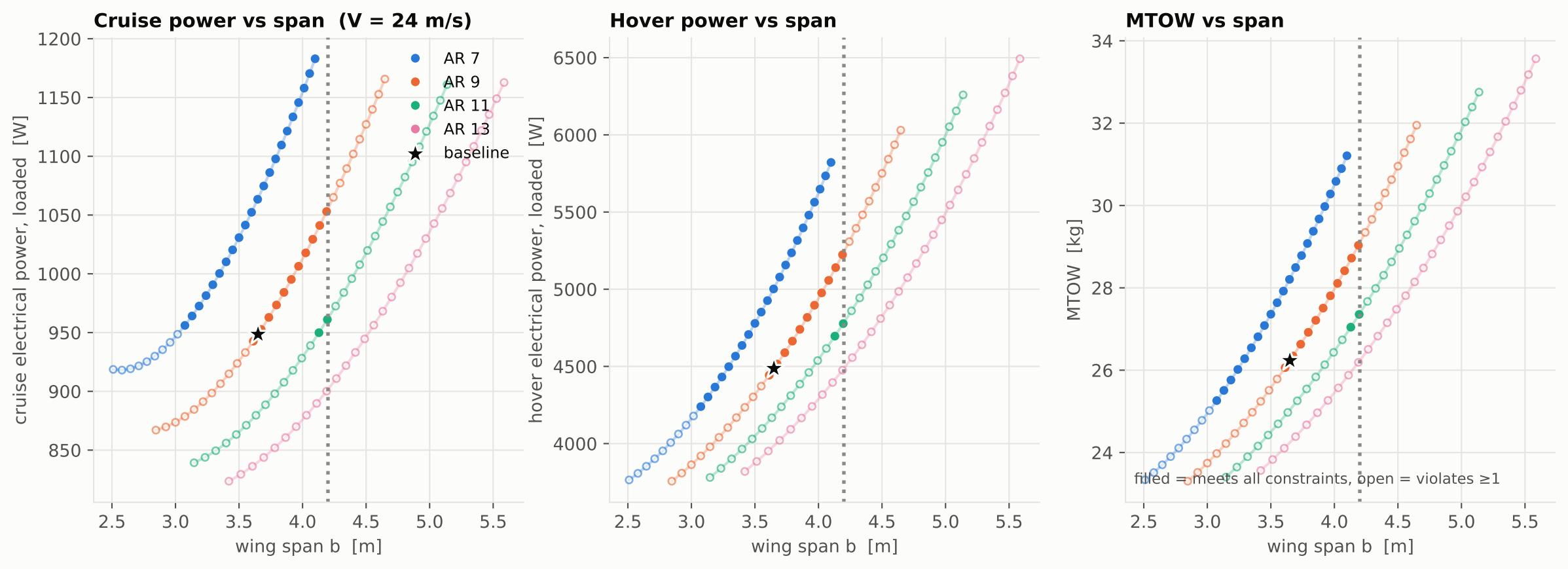
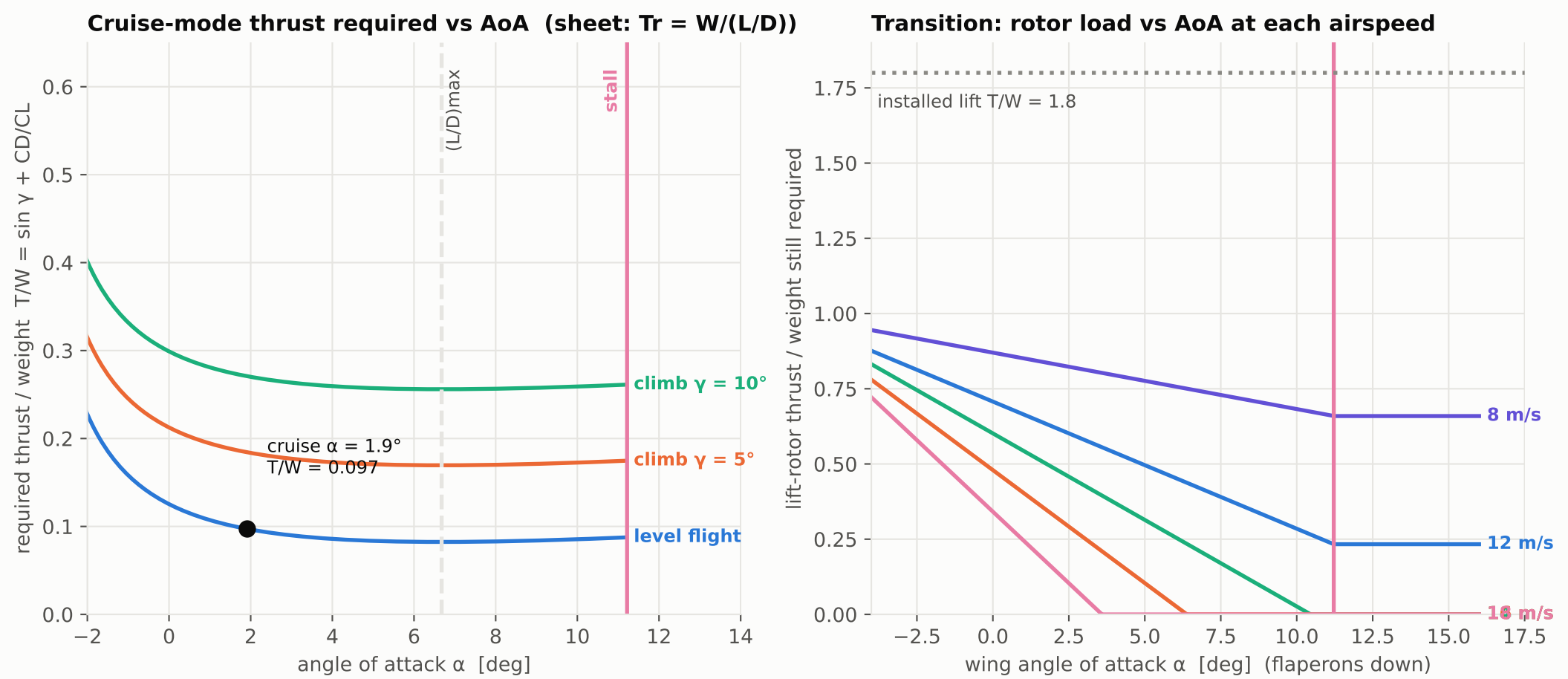
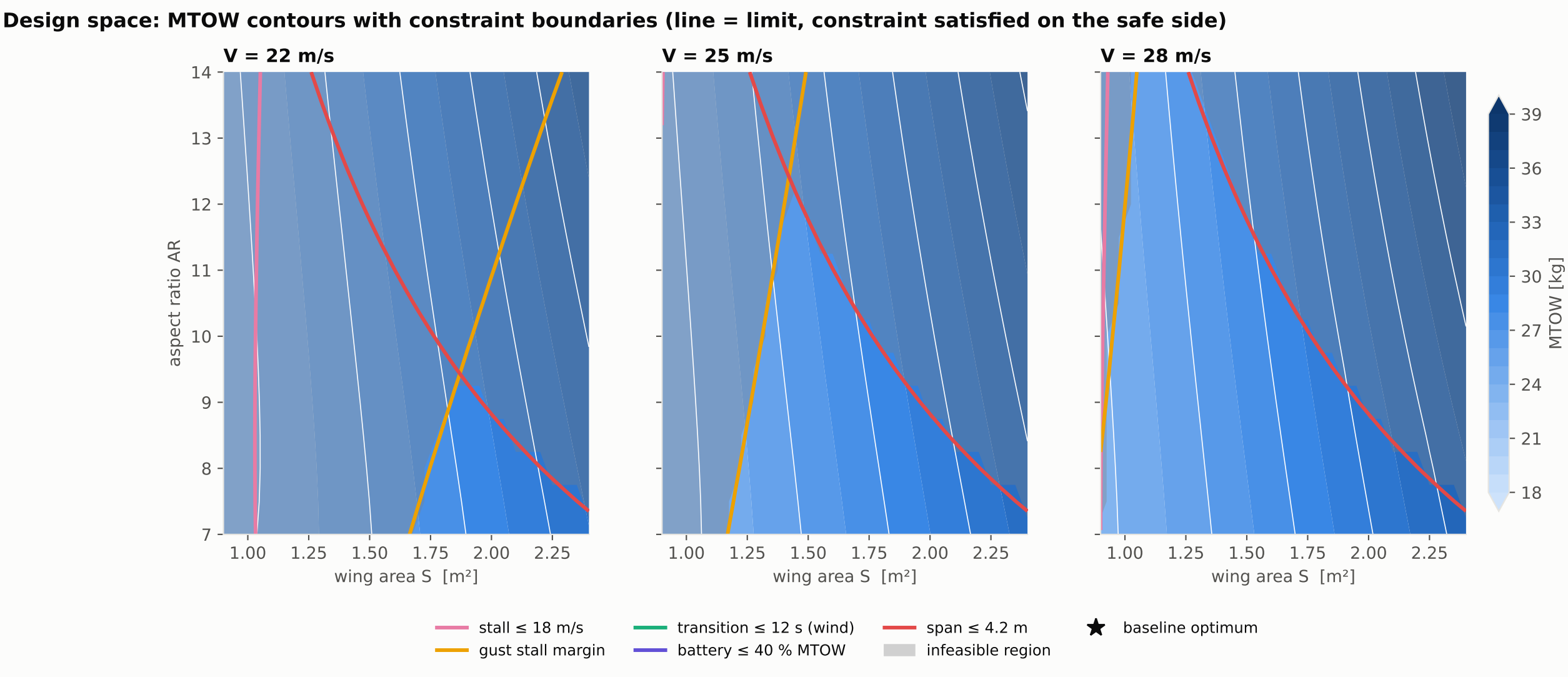
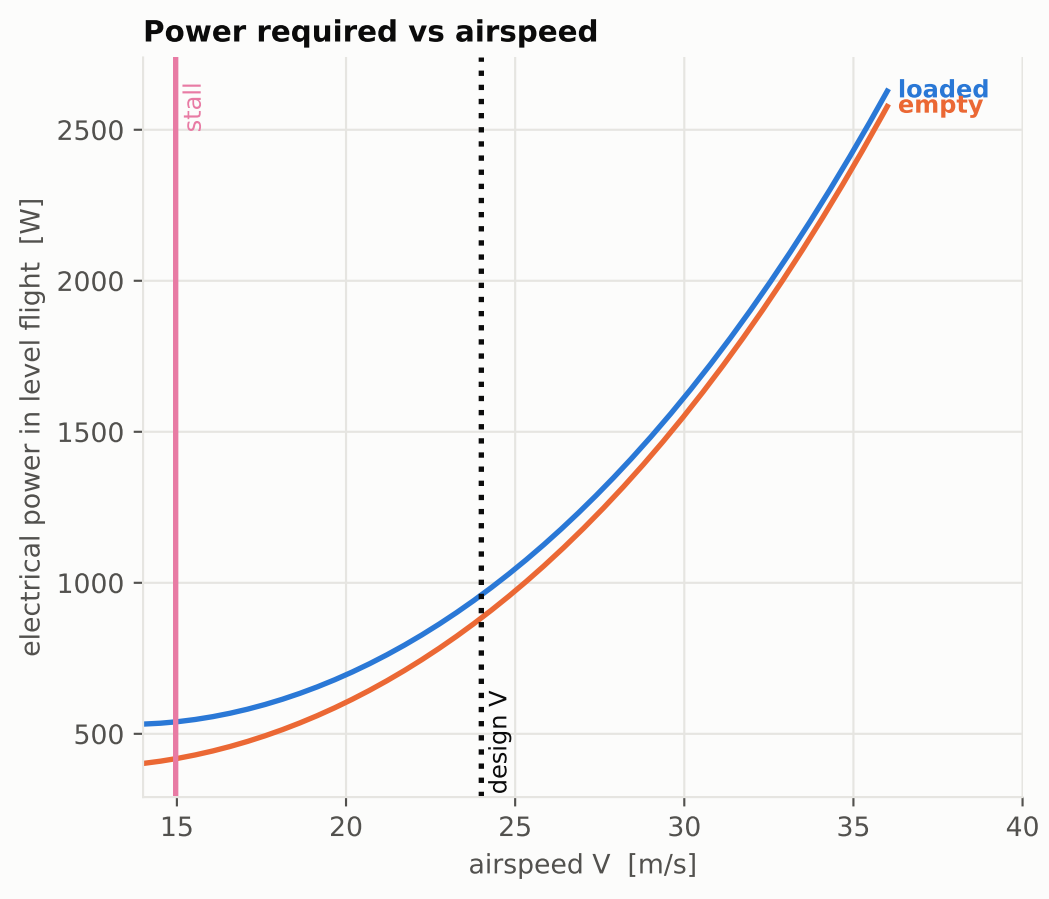
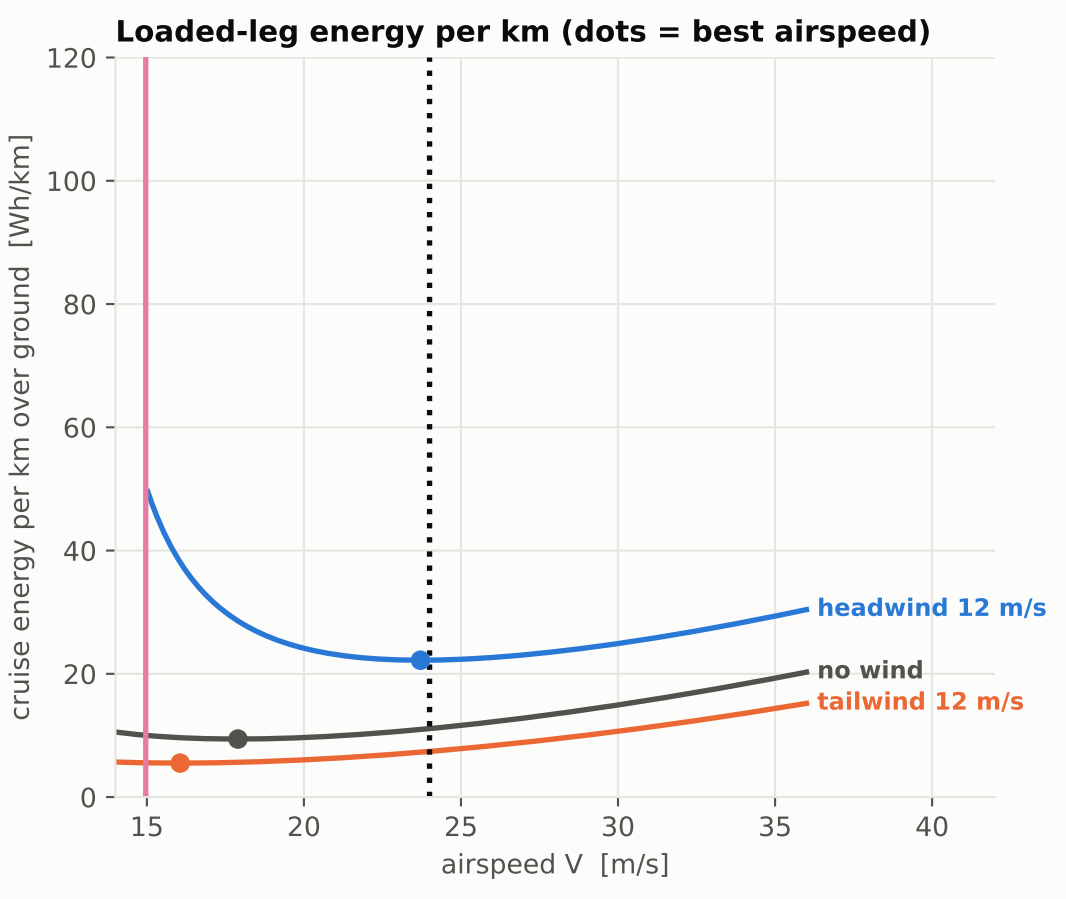

Pusher matching chart (MTOW 26.9 kg, V = 24 m/s)
feasible
infeasible
g'tt 5 m/s
stall 8 m/s
transition (static T/W 0.25)
climb 2 m/s
cruise x1.3
cruise
baseline design point
robust design point
installed pusher power / MTOW [W/kg]
wing loading W/S [N/m²]
Hover: power vs disc loading
design (28 in x4)
FM 0.50
FM 0.60
FM 0.70
hover electrical power / MTOW [W/kg]
disc loading T/A [N/m²]

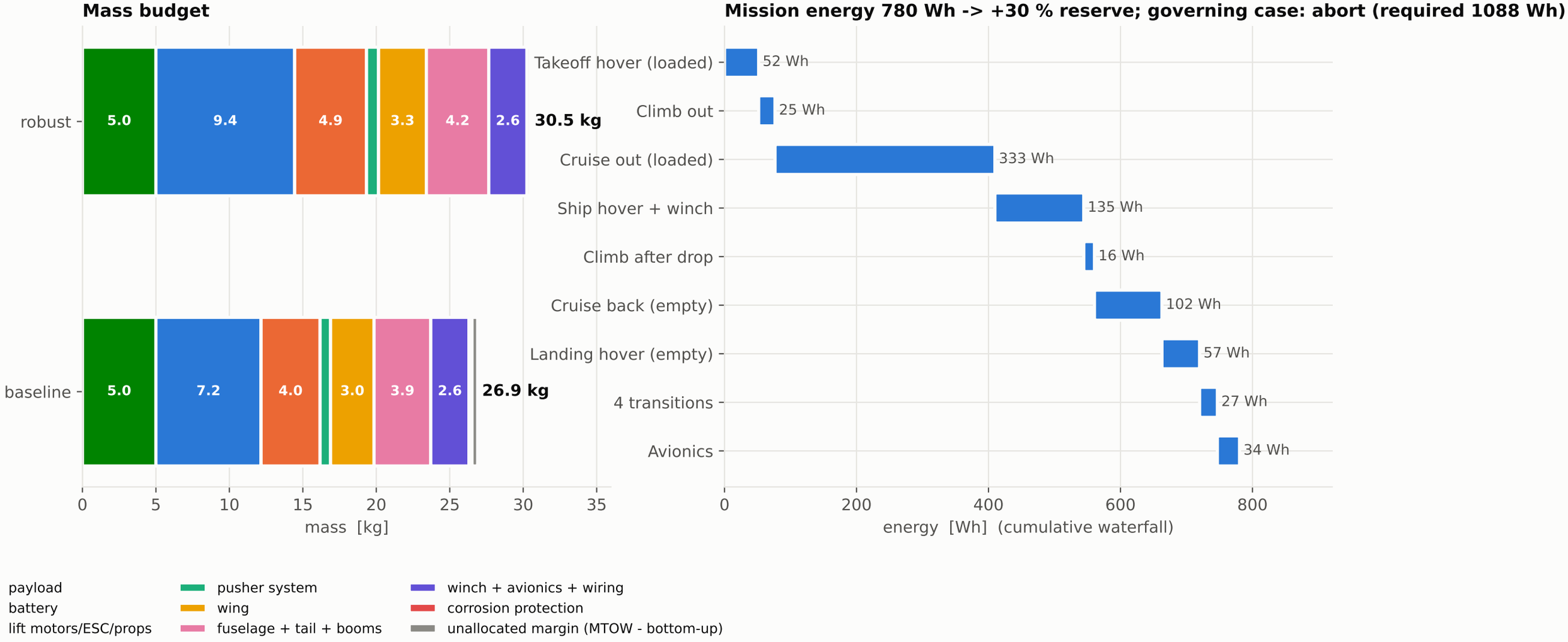

Cruise power vs span (V = 24 m/s)
cruise electrical power, loaded [W]
wing span b [m]
AR 7
AR 9
AR 11
AR 13
baseline
Hover power vs span
hover electrical power, loaded [W]
wing span b [m]
MTOW vs span
MTOW [kg]
wing span b [m]
filled = meets all constraints, open = violates ≥1

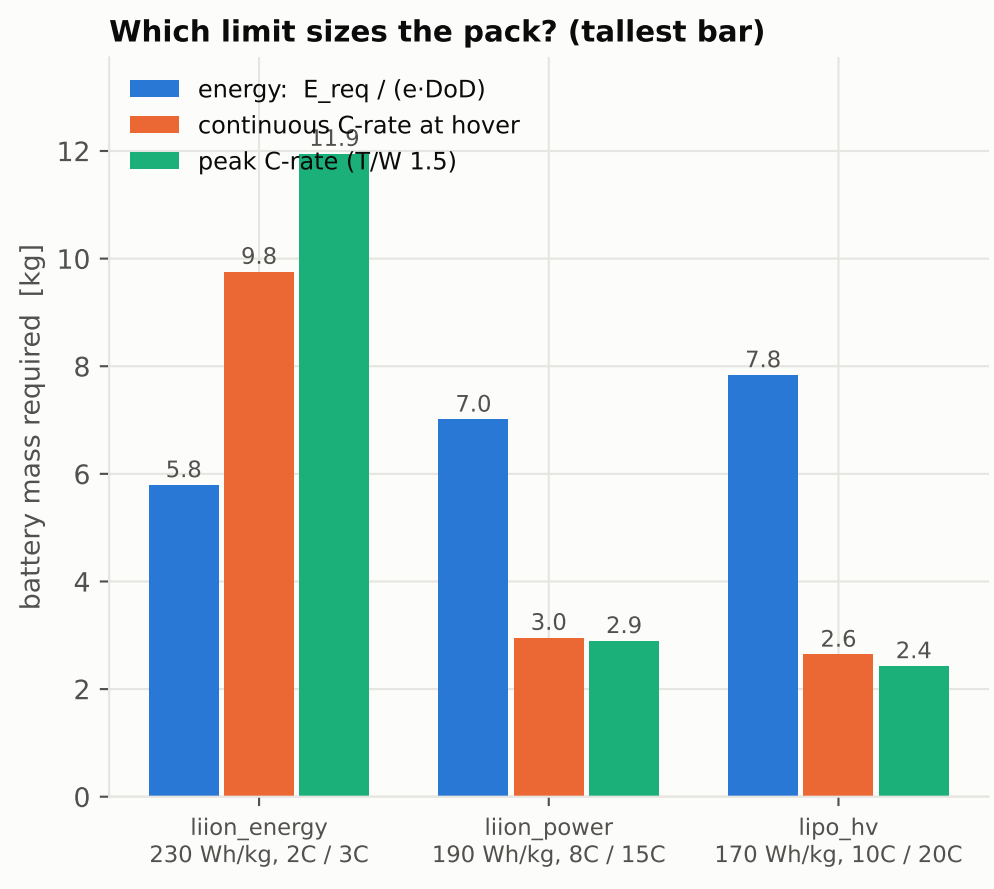

Cruise-mode thrust required vs AoA (sheet: Tr = W/(L/D))
required thrust / weight T/W = sin γ + CD/CL
angle of attack α [deg]
(L/D)max
stall
climb γ = 10°
climb γ = 5°
level flight
cruise α = 1.9°
T/W = 0.097
Transition: rotor load vs AoA at each airspeed
lift-rotor thrust / weight still required
wing angle of attack α [deg] (flaperons down)
installed lift T/W = 1.8
8 m/s
12 m/s
16 m/s

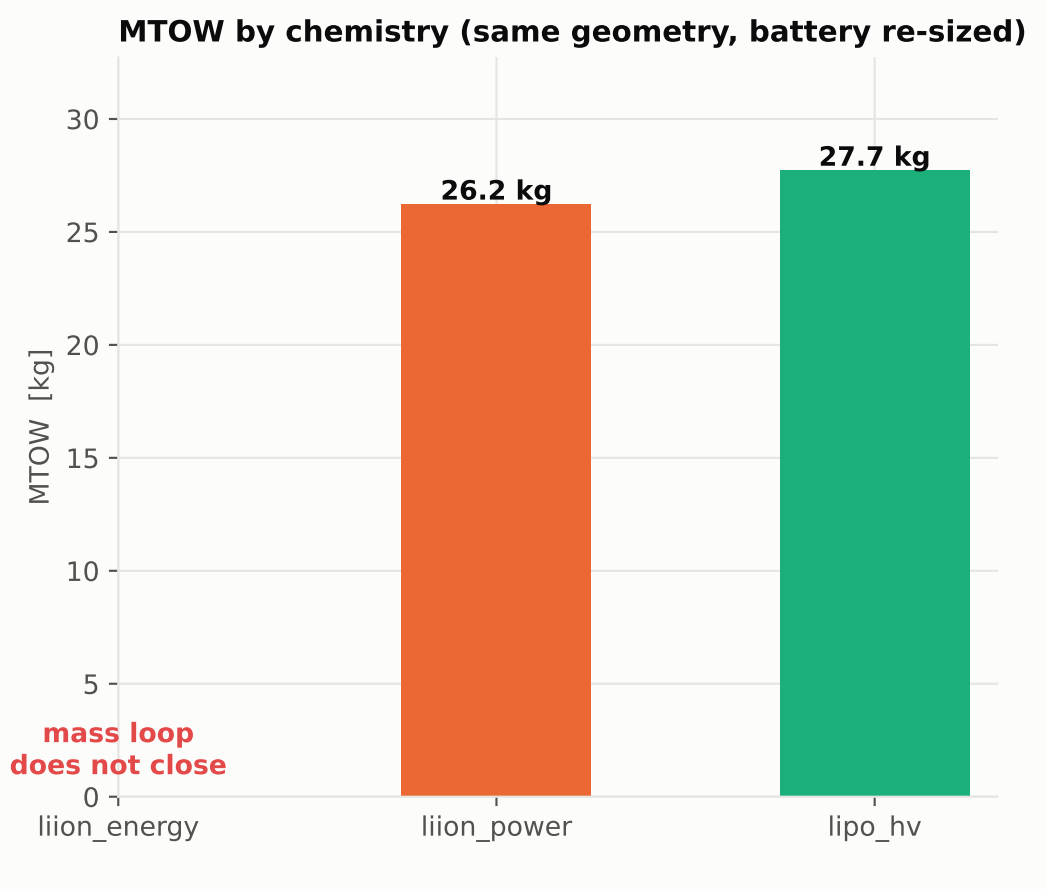

Design space: MTOW contours with constraint boundaries (line = limit, constraint satisfied on the safe side)
V = 22 m/s
V = 25 m/s
V = 28 m/s
aspect ratio AR
14
13
12
11
10
9
8
7
1.00
1.25
1.50
1.75
2.00
2.25
wing area S [m²]
MTOW [kg]
39
36
33
30
27
24
21
18
stall ≤ 18 m/s
gust stall margin
transition ≤ 12 s (wind)
battery ≤ 40 % MTOW
span ≤ 4.2 m
infeasible region
baseline optimum

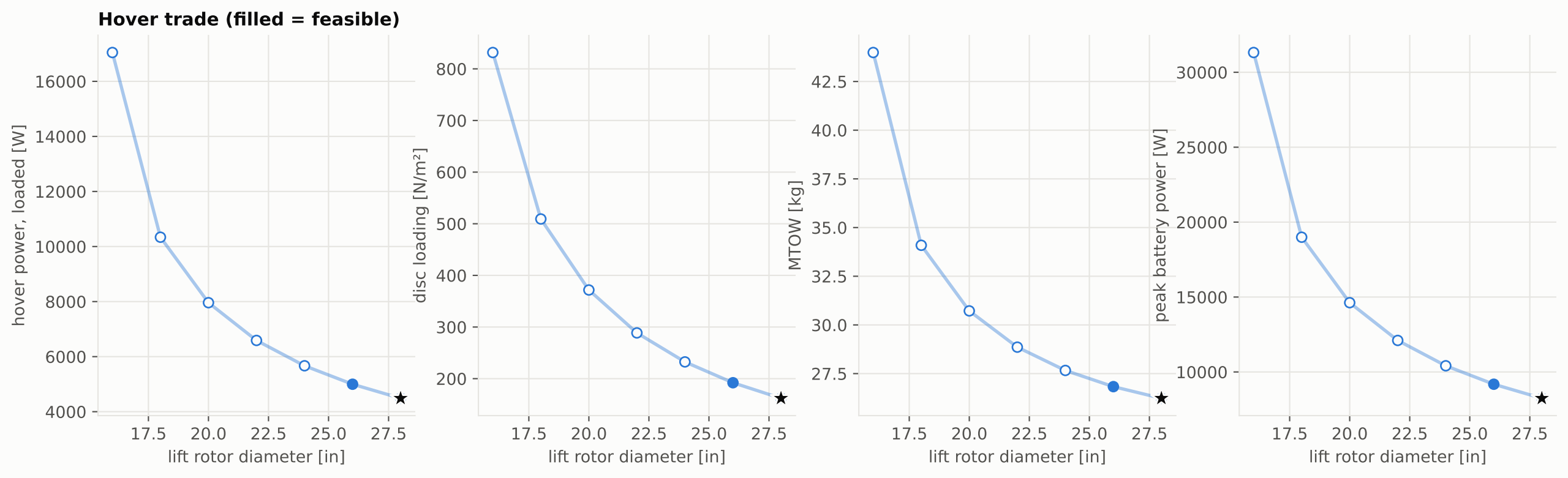

Power required vs airspeed
stall
design V
loaded
empty
electrical power in level flight [W]
500
1000
1500
2000
2500
15
20
25
30
35
40
airspeed V [m/s]


Loaded-leg energy per km (dots = best airspeed)
cruise energy per km over ground [Wh/km]
0
20
40
60
80
100
120
headwind 12 m/s
no wind
tailwind 12 m/s
15
20
25
30
35
40
airspeed V [m/s]

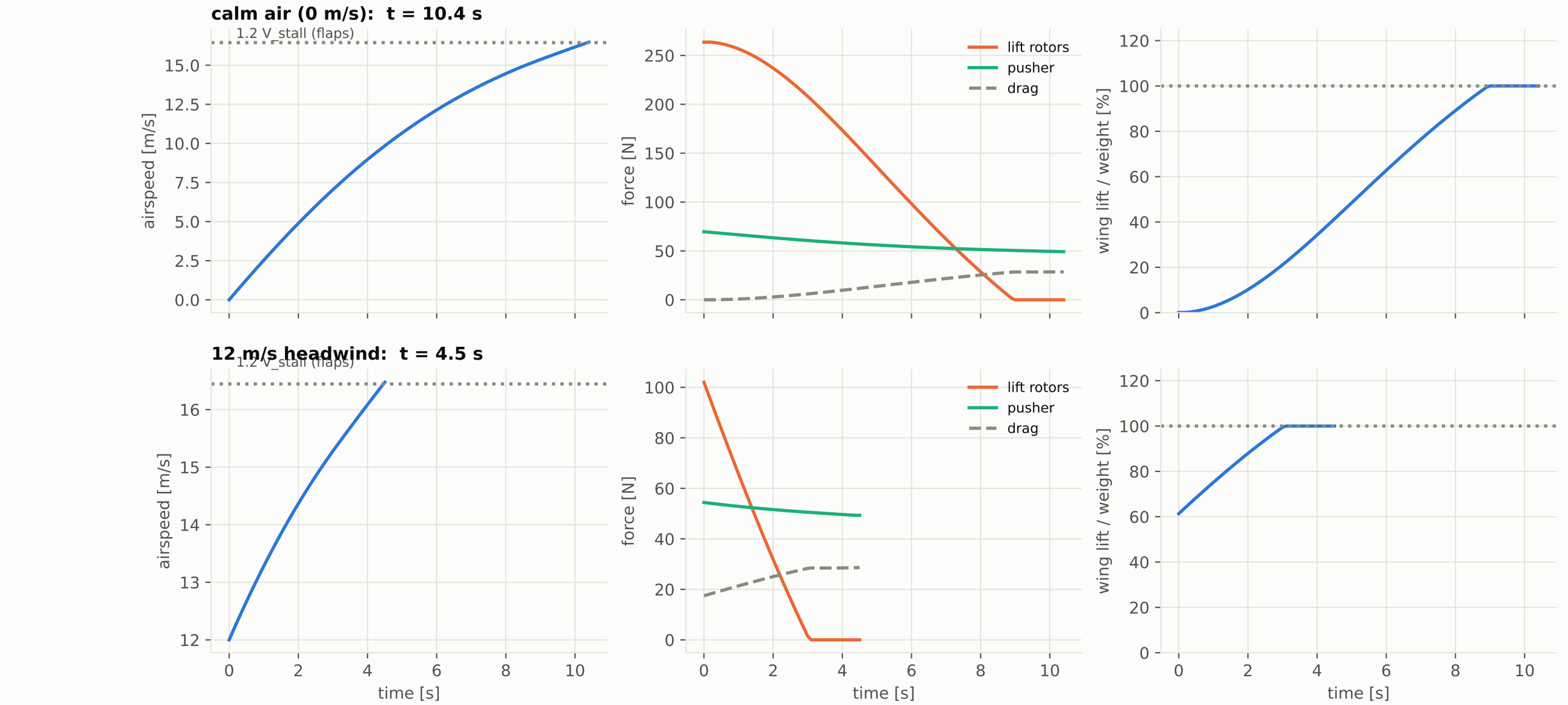

Mass budget
robust
5.0
9.4
4.9
3.3
4.2
2.6
30.5 kg
baseline
5.0
7.2
4.0
3.0
3.9
2.6
26.9 kg
0
5
10
15
20
25
30
35
mass [kg]
Mission energy 780 Wh -> +30 % reserve; governing case: abort (required 1088 Wh)
Takeoff hover (loaded)
52 Wh
Climb out
25 Wh
Cruise out (loaded)
333 Wh
Ship hover + winch
135 Wh
Climb after drop
16 Wh
Cruise back (empty)
102 Wh
Landing hover (empty)
57 Wh
4 transitions
27 Wh
Avionics
34 Wh
0
200
400
600
800
energy [Wh] (cumulative waterfall)
payload
battery
lift motors/ESC/props
pusher system
wing
fuselage + tail + booms
winch + avionics + wiring
corrosion protection
unallocated margin (MTOW - bottom-up)

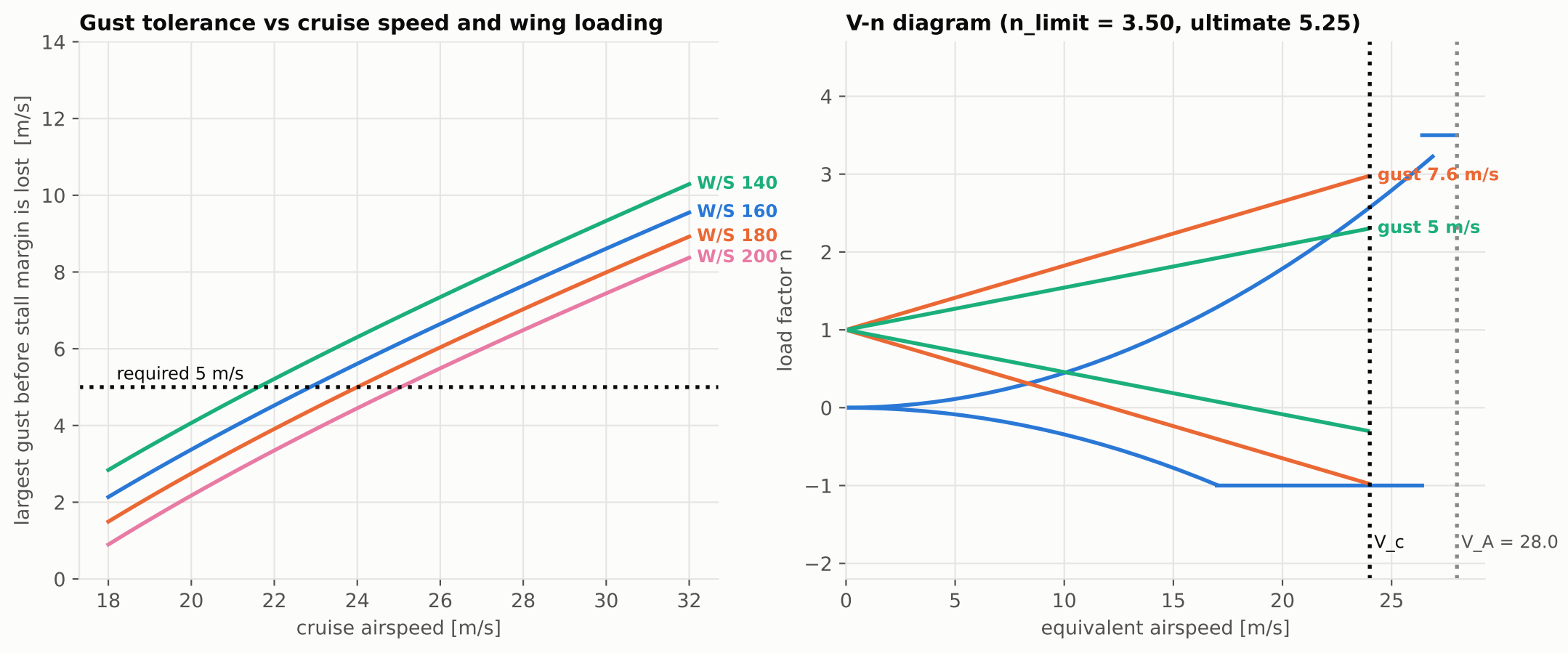

Which limit sizes the pack? (tallest bar)
energy: E_req / (e·DoD)
continuous C-rate at hover
peak C-rate (T/W 1.5)
battery mass required [kg]
0
2
4
6
8
10
12
5.8
9.8
11.9
7.0
3.0
2.9
7.8
2.6
2.4
liion_energy
230 Wh/kg, 2C / 3C
liion_power
190 Wh/kg, 8C / 15C
lipo_hv
170 Wh/kg, 10C / 20C

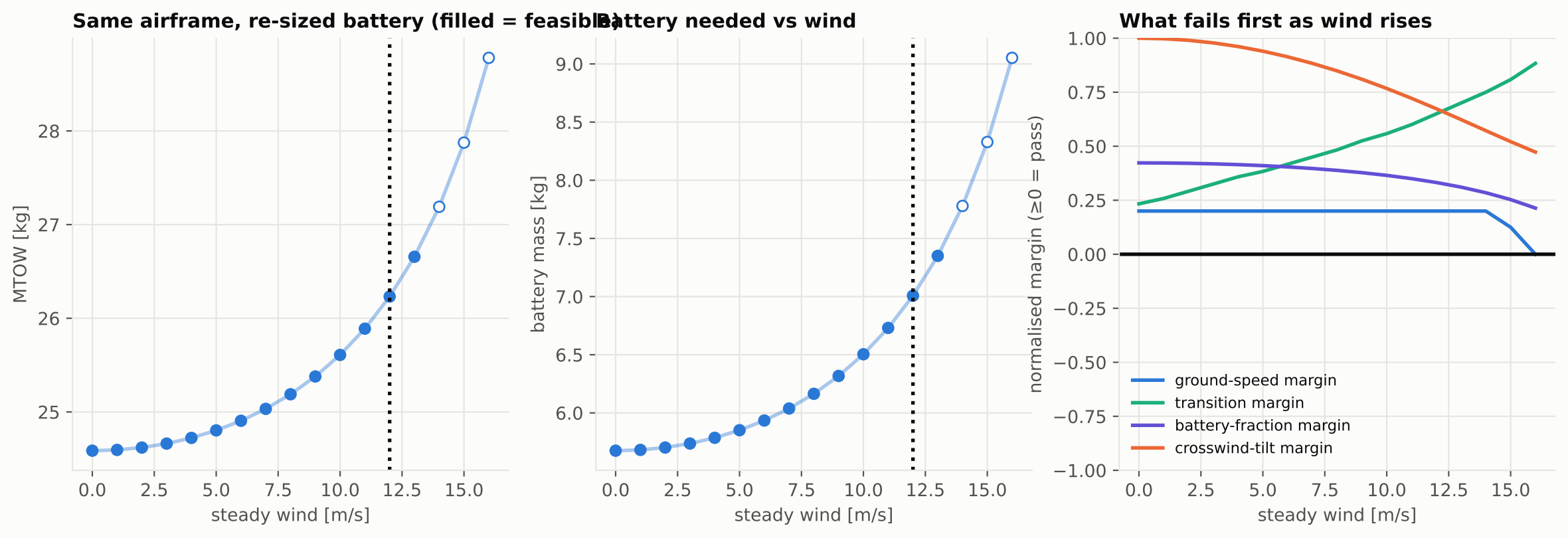

MTOW by chemistry (same geometry, battery re-sized)
MTOW [kg]
0
5
10
15
20
25
30
mass loop
does not close
26.2 kg
27.7 kg
liion_energy
liion_power
lipo_hv

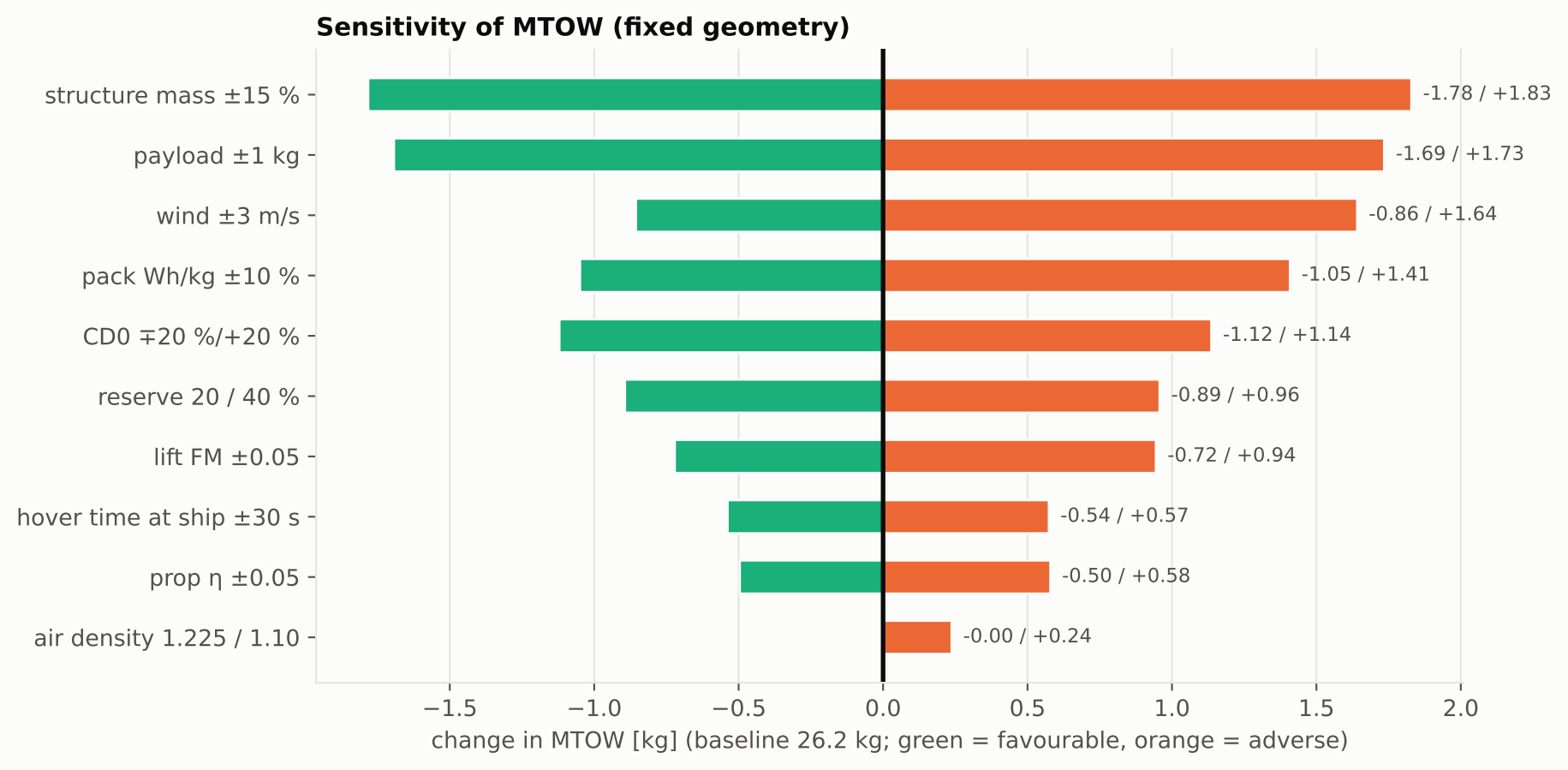

Hover trade (filled = feasible)
hover power, loaded [W]
16000
14000
12000
10000
8000
6000
4000
17.5
20.0
22.5
25.0
27.5
lift rotor diameter [in]
disc loading [N/m²]
800
700
600
500
400
300
200
17.5
20.0
22.5
25.0
27.5
lift rotor diameter [in]
MTOW [kg]
42.5
40.0
37.5
35.0
32.5
30.0
27.5
17.5
20.0
22.5
25.0
27.5
lift rotor diameter [in]
peak battery power [W]
30000
25000
20000
15000
10000
17.5
20.0
22.5
25.0
27.5
lift rotor diameter [in]

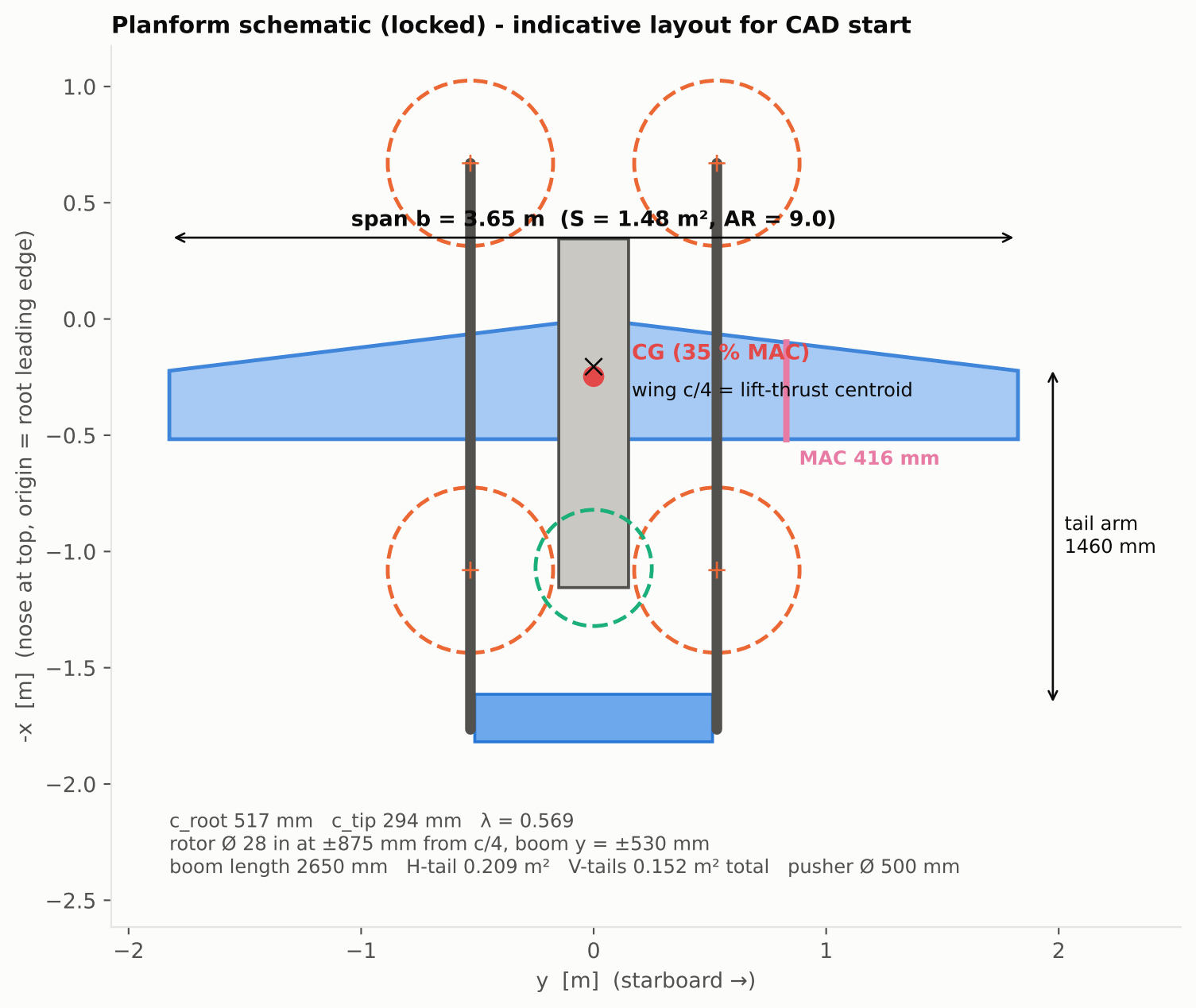
Forward transition (pusher at full power, lift rotors fade out as the wing takes the load)




Gust tolerance vs cruise speed and wing loading
largest gust before stall margin is lost [m/s]
cruise airspeed [m/s]
required 5 m/s
W/S 140
W/S 160
W/S 180
W/S 200
V-n diagram (n_limit = 3.50, ultimate 5.25)
load factor n
equivalent airspeed [m/s]
gust 7.6 m/s
gust 5 m/s
V_c
V_A = 28.0


Same airframe, re-sized battery (filled = feasible)
MTOW [kg]
steady wind [m/s]
Battery needed vs wind
battery mass [kg]
steady wind [m/s]
What fails first as wind rises
normalised margin (≥0 = pass)
steady wind [m/s]
ground-speed margin
transition margin
battery-fraction margin
crosswind-tilt margin


Sensitivity of MTOW (fixed geometry)
structure mass ±15 %
payload ±1 kg
wind ±3 m/s
pack Wh/kg ±10 %
CD0 ∓20 %/+20 %
reserve 20 / 40 %
lift FM ±0.05
hover time at ship ±30 s
prop η ±0.05
air density 1.225 / 1.10
-1.78 / +1.83
-1.69 / +1.73
-0.86 / +1.64
-1.05 / +1.41
-1.12 / +1.14
-0.89 / +0.96
-0.72 / +0.94
-0.54 / +0.57
-0.50 / +0.58
-0.00 / +0.24
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
change in MTOW [kg] (baseline 26.2 kg; green = favourable, orange = adverse)


Planform schematic (locked) - indicative layout for CAD start
span b = 3.65 m (S = 1.48 m², AR = 9.0)
CG (35 % MAC)
wing c/4 = lift-thrust centroid
MAC 416 mm
tail arm
1460 mm
c_root 517 mm c_tip 294 mm λ = 0.569
rotor Ø 28 in at ±875 mm from c/4, boom y = ±530 mm
boom length 2650 mm H-tail 0.209 m² V-tails 0.152 m² total pusher Ø 500 mm
-x [m] (nose at top, origin = root leading edge)
y [m] (starboard →)
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−2
−1
0
1
2